WEEK 1 · DAY 2 · RULE BOOK

Charts That Don't Lie

Follow it step by step. Every common error and its fix is here.

Your goal today: catch the daily *average* hiding a month of outages, compare six zones fairly, and write a title and a 4-sentence finding. Earn **14 green ✅ ticks**. Time: about 60 minutes. Keep this page open next to Colab.

Before you start (2 minutes)

- ☐ Open **W1D2-exercise** in Colab → **File** → **Save a copy in Drive**. Work in the *Copy of…* tab.
- ☐ Run **STEP 0** (Shift + Enter). Wait for ✅ `Ready!`
- ☐ Run every cell **top to bottom**. Replace every `___`, run, read the ✅ or ❌.

The 5 honest-chart rules (today's big idea)

#	Rule	In code
1	**Draw the threshold** (the line that matters: 0 dB, the SLA %)	`plt.axhline(0, color="red", ls="--")`
2	**Title = the finding**, with a number	`plt.title("Link went down on 11 of 31 days")`
3	**Sensible axis**: counts start at 0; bounded % near the top may zoom in, **say the range**	`plt.ylim(0, None)` or `plt.ylim(96, 100)`
4	**One y-axis per chart.** Two units? Two stacked charts	`plt.subplots(2, 1, sharex=True)`
5	**Units on the axis**	`plt.ylabel("link margin (dB)")`

Pick the right daily summary

Your question	Use	Why
"Did it ever go down?" / SLA breach	`.resample("D").min()`	the **worst moment** of each day
"Did it ever overheat?"	`.resample("D").max()`	the worst moment, other direction
"What's the typical level?" / trend	`.resample("D").mean()`	averages smooth away short events

The exercise, step by step

Task	What you do	The tool you need
A1	Frequency code for hourly	`"h"` (lowercase, in quotes)
A2	Count outage hours	`(rf["link_margin_db"] < 0).sum()`
A3	Lowest daily average	`daily_mean.min()`
A4	Days with an outage, from daily minimums	`.resample("D").min()`
A5	Threshold at 0 + a title with a number	`THRESHOLD = 0`, `my_title = "…"`
A6	Which summary for an SLA question?	`"min"`
A7	Uplink power: cause or response?	`"response"`
B1	Pivot long → wide	`"week_start"`, `"region"`, `"uptime_pct"`
B2	Small multiples on the same y-axis?	`True` (no quotes)
B3	Lowest zone on average	`wide.mean().idxmin()`
B4	Zone that fell furthest from its own week 1	`indexed.min().idxmin()`
B5	Axis for tickets? for availability?	`"zero"`, `"tight"`
B6	Order of the 4-sentence finding	Situation, Observation, Implication, Recommendation
🌶️ **Bonus**	Lightest rain that caused an outage	`rf.loc[condition, "rain_rate_mmhr"].min()`

Recipes you'll reuse

```
# long -> wide (one column per group)
wide = df.pivot(index="week_start", columns="region", values="uptime_pct")

# small multiples on ONE shared y-axis
wide.plot(subplots=True, layout=(2, 3), sharey=True, figsize=(12, 5), legend=False)

# index every group to its own first value (= 100)
indexed = wide / wide.iloc[0] * 100
```

The 4-sentence finding

1. **Situation**: what the unit does / the context.
2. **Observation**: what the data showed, **with a number**.
3. **Implication**: why it matters.
4. **Recommendation**: what to do next. **Always last.**

When things go wrong: error → fix

You see	It means	Fix
`ValueError: Invalid frequency: H`	Old uppercase code	Use lowercase `"h"`
`AttributeError: 'Resampler' object has no attribute '___'`	The blank in A4 is still there	Type the method name: `min`
`KeyError: 'region'`	Column name typo	Copy names exactly from `vsat.head()`
`ValueError: Index contains duplicate entries, cannot reshape`	Wrong columns in `pivot`	index = the week, columns = the zone
`NameError: name 'daily_min' is not defined`	You skipped A4	Run A3 and A4 first
`AssertionError: ✍️ Do … first`	A blank above is still empty	Go back, fill it, run again
Six tiny charts look equally bad	`sharey=False`	Use `sharey=True`
Everything is weird	–	**Runtime → Restart session and run all**

Stuck? Climb the ladder

1. Read the ❌ hint. 2. Check this rule book. 3. Ask your neighbour. 4. Ask the AI button to *explain the error* (code only, never real data). 5. Raise your hand.

🏁 **Finished?** Run `scoreboard()` . 14 of 14 = 🏆 . Then write your 4-sentence finding about the six zones. Your trainer will pick a few to read out. 🎤